Activity 3.1: Proving Triangles Congruent

A

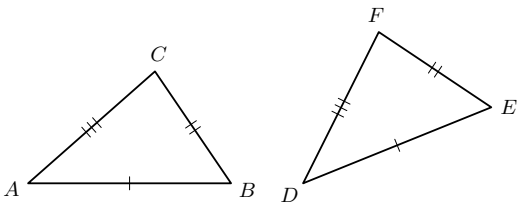
Name: _____ Partner's name: _____ Date: _____

Directions: For #1–4, name the rule that proves the triangles congruent (SSS, SAS, ASA, AAS, or HL). For #5, explain. For #6–8, find the value. Tick and arc marks show congruent parts.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

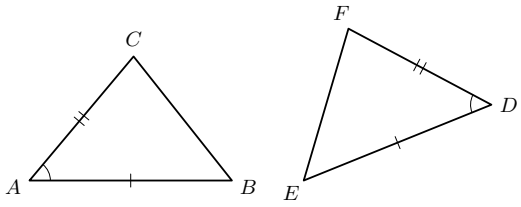
Which rule proves $\triangle ABC \cong \triangle DEF$?



Answer: _____ Match? ☐ Yes ☐ No

2

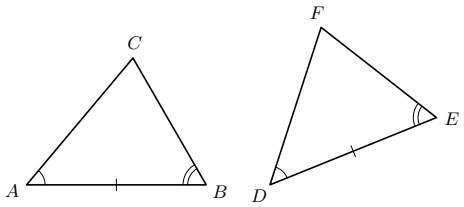
Which rule proves $\triangle ABC \cong \triangle DEF$?



Answer: _____ Match? ☐ Yes ☐ No

3

Which rule proves $\triangle ABC \cong \triangle DEF$?



Answer: _____ Match? ☐ Yes ☐ No

4

$\triangle ABC$ and $\triangle DEF$ are right triangles with right angles at C and F. $\overline{AB} \cong \overline{DE}$ and $\overline{AC} \cong \overline{DF}$. Which rule proves them congruent?

Answer: _____ Match? ☐ Yes ☐ No

5

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Can you conclude $\triangle ABC \cong \triangle DEF$? Explain.

Answer: _____ Match? ☐ Yes ☐ No

6

$\triangle ABC \cong \triangle DEF$, $AB = 3x + 2$, and $DE = 5x - 6$. Find AB.

Answer: _____ Match? ☐ Yes ☐ No

7

$\triangle ABC \cong \triangle DEF$, $m\angle A = (4x + 8)°$, and $m\angle D = (6x - 12)°$. Find $m\angle A$.

Answer: _____ Match? ☐ Yes ☐ No

8

$\triangle ABC \cong \triangle DEF$ with $AB = 5$, $BC = 7$, and $CA = 9$. Find EF.

Answer: _____ Match? ☐ Yes ☐ No

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B

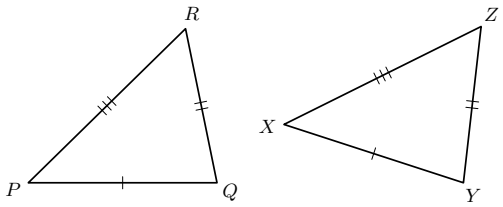
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1

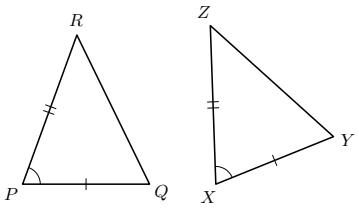
Which rule proves $\triangle PQR \cong \triangle XYZ$?



Answer: _____ Match? ☐ Yes ☐ No

2

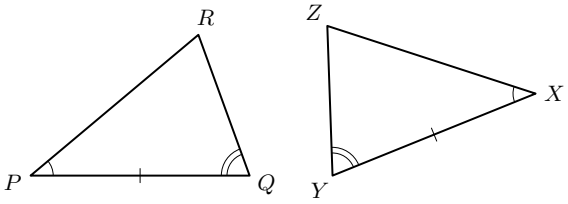
Which rule proves $\triangle PQR \cong \triangle XYZ$?



Answer: _____ Match? ☐ Yes ☐ No

3

Which rule proves $\triangle PQR \cong \triangle XYZ$?



Answer: _____ Match? ☐ Yes ☐ No

4

$\triangle PQR$ and $\triangle XYZ$ are right triangles with right angles at R and Z. $\overline{PQ} \cong \overline{XY}$ and $\overline{QR} \cong \overline{YZ}$. Which rule proves them congruent?

Answer: _____ Match? ☐ Yes ☐ No

5

Two triangles have three pairs of congruent angles, but no side lengths are known. Can you conclude the triangles are congruent? Explain.

Answer: _____ Match? ☐ Yes ☐ No

6

$\triangle PQR \cong \triangle XYZ$, $PQ = 2x+4$, and $XY = 4x-6$. Find PQ.

Answer: _____ Match? ☐ Yes ☐ No

7

$\triangle LMN \cong \triangle RST$, $m\angle M = (3x+18)°$, and $m\angle S = (5x-2)°$. Find $m\angle M$.

Answer: _____ Match? ☐ Yes ☐ No

8

$\triangle PQR \cong \triangle XYZ$ with $PQ = 5$, $RP = 9$, and $QR = 7$. Find YZ.

Answer: _____ Match? ☐ Yes ☐ No